ShadowNav - AA272 Final Project

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ABSTRACT

With dramatic improvements in space launch capabilities in the past two decades, the Moon is now a focus for both crewed and robotic missions from around the globe. One significant challenge with autonomous operations on the Moon is reliable localization. While Earth-centered GNSS can be leveraged to provide some localization, obstructions to the signal will pose a challenge, especially in polar craters where early exploration is likely to take place. In this project, the team created tooling to help design missions to the moon, including ray tracing tools to determine satellite visibility, error estimates for specific potential landing sites, and analysis regarding optimal lunar augmentations to the existing GNSS system. With greater computational resources and further development, this tooling could be used to fully characterize the localization performance that robotic explorers should expect in different parts of the lunar surface.

I. INTRODUCTION

The future Artemis activities will take place near the Moon's south pole, including inside craters that stay in permanent shadow. In these areas, it's still unclear how reliably GNSS signals can reach the surface. In our project, we first examine how much of the GNSS constellation is actually visible inside Shackleton crater, and then look at how a lunar-orbiting augmentation satellite could help fill in those coverage gaps. We begin by constructing the local horizon at Shackleton crater using DEM-based ray tracing, then use this geometry to model GNSS visibility and quantify where and when signals are blocked. Finally, we simulate several candidate polar orbits to understand the relationship between altitude and visibility and examine how often a dedicated lunar satellite would appear above the crater rim. Together, these steps provide a clearer picture of navigation reliability at the south pole and how orbital augmentation could support future surface operations.

II. RELATED WORK

Navigation in obstructed environments has been studied extensively on Earth, and several of those ideas motivate our approach. Groves' work on shadow matching showed that using the pattern of blocked sky regions can improve positioning in urban canyons Groves (2011). This idea parallels our use of crater geometry to reason about GNSS visibility inside Shackleton. Prior studies have also examined how Earth-GNSS signals propagate to the Moon; Delépaut et al. demonstrated that these signals do reach the lunar surface, but they are extremely weak and often geometrically clustered near the poles Delépaut et al. (2021). Work on proposed lunar GNSS constellations similarly shows that dilution of precision degrades at high latitudes, motivating site-specific analysis for south-pole operations Pereira et al. (2022).

For terrain modeling, we rely on LOLA DEM data from NASA's PGDA, which is commonly used in lunar topography studies Smith et al. (2016). Our orbital modeling uses standard two-body propagation and reference-frame transformations provided by poliastro pol (2024a) and local ENU conversions via pymap3d Greenspan and contributors (2023). These tools and prior studies form the foundation for our combined analysis of GNSS visibility and augmentation with a lunar-orbiting satellite.

III. PROBLEM STATEMENT

Precise and reliable positioning on the moon is essential for future space missions, robotic exploration, obtaining resources, and sample-return operations. Several regions of scientific and operational interest, including Gruithuisen Domes, Marius Hills, Rima Bode, and the permanently shadowed Shackleton Crater, have really complex topography that significantly challenges conventional navigation methods.

Unlike Earth, the Moon can't rely on terrestrial GNSS signals, which arrive at extremely low power levels and do not provide usable satellite geometry. Onboard sensors such as inertial measurement units (IMUs) accumulate drift over time, and vision-based navigation degrades or becomes entirely unavailable in low-illumination environments, particularly within permanently shadowed regions (PSRs). Furthermore, lunar terrain features such as crater rims, volcanic domes, and rille walls block line-of-sight to orbiting satellites, leading to intermittent visibility, poor geometry, and degraded Dilution of Precision (DOP).

These lunar terrain features have effects similar to Urban Canyons on Earth, which require further exploration and analysis to determine how bad they are and which solutions could be proposed. By analyzing the geometry of different sites of the moon and using these geometries to model elevation masks, initial estimations of DOPs can be made, and then improved by different methods.

By combining GNSS theory, DOP analysis, and site-specific terrain masking, this work characterizes the navigation challenges on the lunar surface and establishes the performance criteria necessary for the a Moon Navigation System Architecture that takes into account terrain features.

IV. RAY TRACING SATELLITE VISIBILITY ANALYSIS

1. Approach

Ray tracing generally describes a programming technique in computer graphics where an image is generated by sending out virtual ”rays” from a virtual ”camera.” The computer then calculates what these rays intersect which leads to the actual image generation. This intersection calculation is the core of ray tracing and must be very fast in order to make ray tracing feasible in a complex scene. Thus, complex shapes need to be converted into a combination of ”primitives” which are simple geometric forms that have a closed form intersection equation. The most common such primitive is a a triangle, which was used in this project to convert the moon shape into a mesh.

The first step in setting up this mesh was simply acquiring lunar elevation data. For this project, this was taken from NASA’s Scientific Visualization Studio. This produced an image like the one seen on the left side of Figure 1 below.

Since this data has a large number of pixels, it was then down-sampled by 12x to produce a dataset that was easier to work with as seen in the right side of Figure 1 below. Because of the two dimensional nature of the dataset, down-sampling by 12x in each direction results in a 144x speedup of the overall runtime.

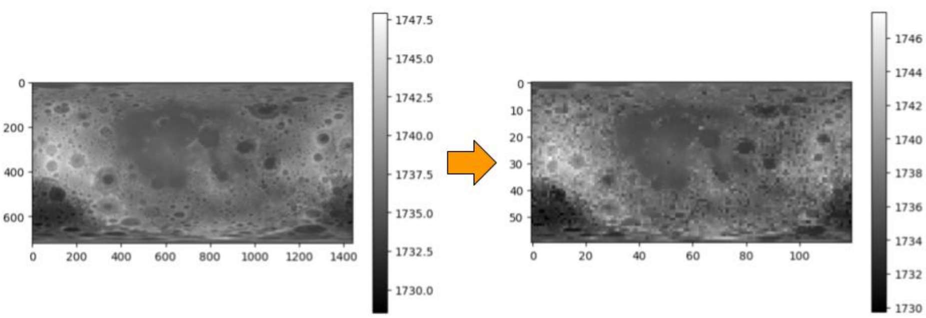


Figure 1: An elevation map (in km from the lunar center) being downsampled

The 2D image had to then be applied to a 3D sphere. To do this, the pixel locations were first converted into latitude and longitude. These two coordinates with the elevation of the point were then used as spherical coordinates to create a series of points in Cartesian space. By connecting these points (along with two points for the North and South poles) into triangles, a mesh could be created for the lunar surface. This mesh can be seen in Figure 2 below.

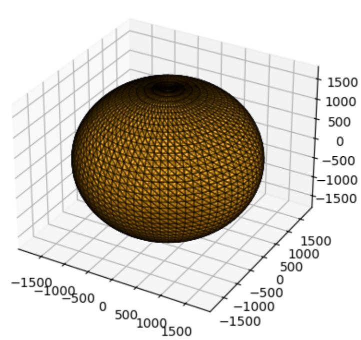


Figure 2: The mesh created from the elevation map (20x downsampling used here for improved visualization

With this mesh created, the only remaining part of the ”scene” to set up was the satellites themselves. Since GNSS satellites focus the majority of their energy down towards the Earth, proposals for using pre-existing GNSS constellations for lunar localization often focus on signals sent just past the Earth. Thus, a band of "satellites" were created in a circle just above the edge of Earth's atmosphere, approximating these signals.

From here, the general approach was to iterate through each point on the mesh and create rays from the point to each of the satellites mentioned above (with a slight shift to avoid any self-intersection). Each of these rays was then tested for intersection against each primitive triangle in the moon mesh to determine if it was obstructed. To generate the horizon line, a binary search was done for a number of different azimuth angles trying to determine the boundary between where intersections begin and end.

Several optimizations, such as a Bounding Volume Hierarchy (BVH), could be used to dramatically accelerate the computation, but those will be left for future development work.

2. Results

With this ray tracing approach, two primary analyses could be conducted. First, a lunar map was created analyzing the degree of visibility that different parts of the surface would have. The product of this analysis can be seen in Figure **??** below.

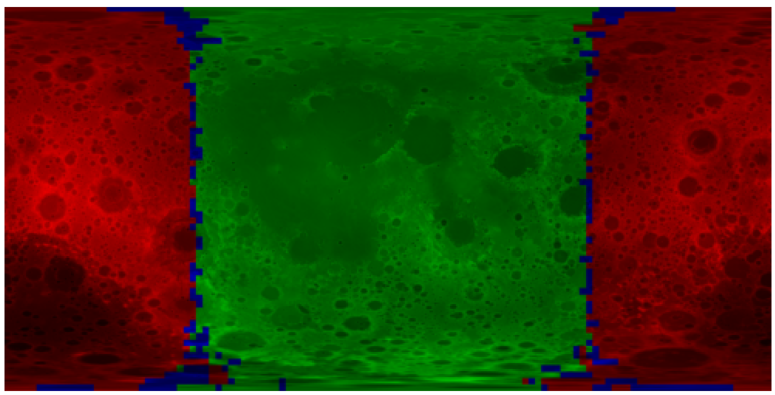


Figure 3: A visibility map of the lunar surface (green = total Earth visibility, blue = partial Earth visibility, red = no Earth visibility)

This map accurately shows a side of the moon facing the Earth and part facing away. Between these green and red regions, blue spots highlight areas where the moon's topography limits the visibility of some of the satellites. Unfortunately, due to the lack of ray tracing optimizations and more advanced computation resources, the resolution is not as high as it could be. This is because this single image took about 8 hours to generate, and it would scale quadratically with increasing numbers of points. With this caveat, this still validates the basic functionality of the tool as it could be used for estimating lunar localization accuracy.

To assess the tooling to determine the horizon line, a sample elevation curve for a crater at the Moon's South Pole can be seen in Figure 4 below.

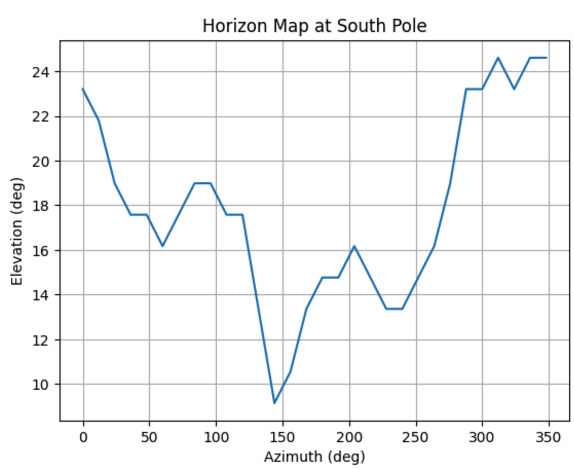


Figure 4: The line showing the elevation at different points along a South pole's craters rim

This graph shows us a rim elevation between 10 and 25 degrees which is to be expected for these lunar craters.

V. SITE ERROR ANALYSIS

After knowing and analyzing the topology of the moon, we could get very precise elevation masks of different sites. In this case, we are particularly interested in sites like craters or volcanic complexes, since they have tall walls that could act as natural Urban Canyons on the moon. To make this project more simple and doable in the time we have, we decided to make two assumptions. First of all, we are going to analyze the craters using a constant elevation for the walls, and second, we are going to use the CSV files that we were given in one of the previous homework's. The reason for our second assumption is that trying to get positing data of GNSS satellites in different parts of the moon involves a much more complex procedure, and although it could be done with hard work and a lot of time, we decided not to do that in this project.

Because of the second assumption, the sites that we will analyze to exemplify and prove our project are Gruithuisen Domes, Marius Hills, Rima Bode and Shackleton. For each of them, we researched and got information of the type of site, diameter, depth, and average elevation angle. Some of the sites are not single craters, but complex volcanic sites or rilles systems, making their geometries and topographies even more complicated. For this project, we decided to simplify them and select a constant elevation assumption (Assumption 1 stated above) for each of the sites just to exemplify what a horizon mask looks like, but this does not represent the real behavior of the site.

- **Gruithuisen Domes**
  Type: Volcanic Domes
  Diameter: 8–33 km
  Height (relief): 1.1–1.8 km
  Average wall/elevation angle: $\approx 10°$
- **Marius Hills**
  Type: Volcanic Complex
  Diameter (Marius Crater): 41 km
  Depth: 1.7 km
  Average wall/elevation angle: $\approx 8°$
- **Rima Bode**
  Type: Rille System
  Diameter (representative feature): 18.6 km
  Depth: 3.5 km
  Average wall/elevation angle: $\approx 15°$
- **Shackleton Crater**
  Type: Impact Crater
  Diameter: 21 km
  Depth: 4.2 km
  Average wall/elevation angle: $\approx 30°$

In the following image, we can see the location of the different sites in the Moon, and we can also see their skyviews with the location of the GNSS satellites on Earth. One important thing to notice is that most of them are in the bright side of the moon pointing towards Earth, but Shackleton is in the South Pole, therefore, the location of the satellites lies in a low elevation zone.

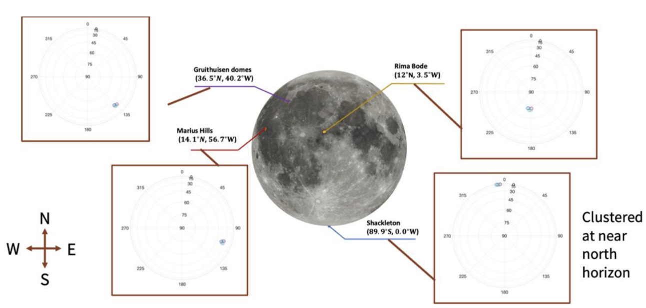


Figure 5: Sites locations on the Moon (HW2, P3, Global Positioning System AA272 Class).

1. Approach

With the information of the elevation mask, and the elevation and azimuth of the satellites in view, we implemented a workflow to evaluate how terrain geometry affects satellite visibility and Dilution of Precision (DOP). Our code performs the following steps:

1. **Load satellite geometry.** For each site, we import the azimuth and elevation of all satellites from the provided CSV files. These represent the open-sky conditions without any terrain blocking.
2. **Generate simplified horizon masks.** Using the average wall elevation angles for each site, we create a $360°$ elevation mask that assigns a constant cutoff angle for every direction. Although simplified, this captures the effect of terrain obstruction in craters or volcanic complexes.
3. **Determine satellite visibility.** For every satellite, we compare its elevation angle with the interpolated horizon mask at the corresponding azimuth. Satellites whose elevation is below the mask are classified as *blocked*, while the rest are labeled as *visible*.
4. **Construct the geometry matrix $\tilde{G}$.** For all visible satellites, we compute the ENU unit direction vectors and form the local geometry matrix as defined in Lecture 4. This matrix captures how the satellite configuration contributes to the estimation of East, North, Up, and Time states.
5. **Compute the DOP matrix.** We evaluate the DOP matrix

$$\tilde{H} = (\tilde{G}^{\top}\tilde{G})^{-1},$$

   and extract PDOP, HDOP, VDOP, and TDOP. These provide a quantitative measure of how terrain masking impacts navigation quality at each site.
6. **Visualize skyplots and DOP matrices.** For clarity, we generate skyplots showing visible and blocked satellites, the horizon mask, and the blocked region. We also visualize the DOP matrix as a heatmap to highlight how the geometry degrades after applying the terrain mask.

This approach allows us to directly compare ideal open-sky geometry with terrain-masked geometry, and to quantify how challenging each site is for satellite-based navigation. Sites with tall crater walls, such as Shackleton, show significant degradation in available satellites and consequently much higher DOP values, while volcanic regions like Marius Hills or Gruithuisen exhibit more moderate geometric loss.

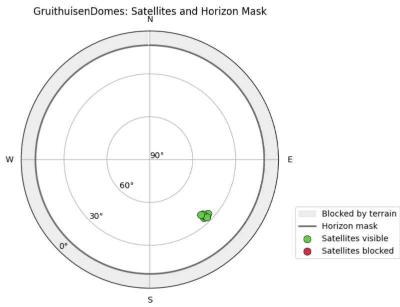
2. Results

Figures 6 and 7 show the satellite visibility skyplots and the corresponding H-matrices for each of the four lunar sites in open skies. These results allow us to quantify how terrain features, modeled here through constant elevation masks, affect satellite geometry and, consequently, Dilution of Precision (DOP). In general, Gruithuisen Domes, Marius Hills and Rima Bode have the satellites in complete view, so the DOP with or without the horizon stays the same. Nevertheless, with Shackleton, the horizon mask completely covers the satellites, so with it, the DOP becomes infinite. Below is a table showing the results for the DOP of all sites in open skies assumptions, and the satellites in view with the horizon mask.

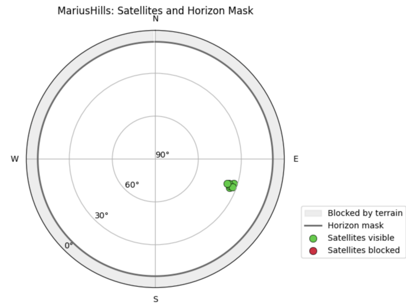
Site	PDOP Open Sky	PDOP Horizon Mask	Satellites in View W Horizon Mask
Gruithuisen Domes	1364.55	1364.55	7
Marius Hills	1364.81	1364.81	7
Rima Bode	1359.58	1359.58	7
Shackleton	1372.15	Inf	0

Table 1: PDOP and visible satellite count for the four analyzed lunar sites.

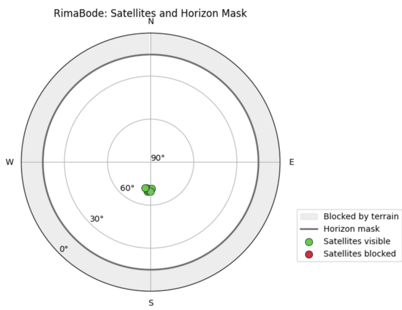
In the following figure, the different satellite skyplots can be seen for the four different locations, with the horizon masks applied. We can see the horizon mask as a gray ring in the perimeter, the higher the elevation the bigger the thickness of the ring. Satellites in red represent non visible satellites, blocked by the horizon mask. Satellites in green represent visible satellites from the site.



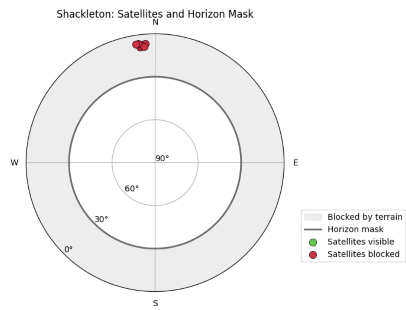
(a) Gruithuisen Domes



(b) Marius Hills



(c) Rima Bode



(d) Shackleton

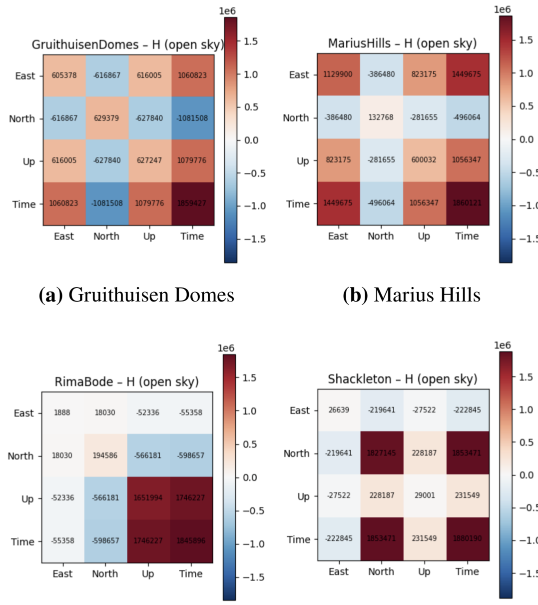
Figure 6: Satellite skyplots and elevation masks for the four analyzed sites.

Gruithuisen Domes. This site exhibits the mildest terrain masking among the four cases. The skyplot shows that all satellites remain visible above the 10° horizon, and only a small portion of the sky is obstructed. Correspondingly, the open-sky H matrix displays moderate cross-coupling between the East, North, and Up components, with no entry dominating the others. This indicates that geometry is relatively well-balanced, producing the lowest overall PDOP among the analyzed sites. As expected for a volcanic dome, the gentle slopes produce only minor degradation in navigation geometry.

Marius Hills. Although the terrain is more complex, the masking effect remains limited due to the relatively low 8° elevation mask used for this site. The skyplot shows that all of satellites are still in view, with obstructions only near the horizon. The H matrix exhibits slightly larger magnitudes compared to Gruithuisen, suggesting slightly poorer geometric strength. However, the diagonal terms remain comparable in scale, indicating that the geometry degrades somewhat uniformly across East, North, and Up. This produces a moderate PDOP that is still acceptable for navigation performance.

Rima Bode. The rille structure introduces a steeper horizon mask of 15°, which is clearly reflected in the skyplot. However, still all of the satellites remain visible. This uneven visibility results in a more anisotropic H matrix, where certain directional terms (particularly Up and Time) show significantly higher magnitude. The geometry becomes more sensitive to vertical and temporal uncertainty, yielding a noticeably higher PDOP.

Shackleton Crater. Shackleton presents the most extreme geometry degradation. With a 30° wall elevation, the skyplot shows that none of the satellites remain visible, all clustered in a narrow azimuthal region and at high elevation. This severely limits the diversity of line-of-sight vectors. The resulting H matrix in open sky has the highest magnitudes across all sites, especially in the Up and Time components. This implies that vertical dilution and clock sensitivity are significantly amplified. The poor geometric spread causes PDOP to increase dramatically, making Shackleton the most challenging environment for GNSS-like navigation. These results closely mirror real-world expectations for deeply shadowed polar craters.



(a) Gruithuisen Domes **(b)** Marius Hills

(c) Rima Bode **(d)** Shackleton

Figure 7: H graphs for the four different sites.

VI. LUNAR SATELLITE AUGMENTATION

To evaluate how a lunar-orbiting satellite could improve navigation inside Shackleton crater, we modeled several candidate polar orbits and measured how often a satellite in each orbit would appear above the crater rim. This analysis involved propagating each orbit, converting satellite positions into the local Shackleton sky frame, and applying the horizon mask generated from the DEM ray-tracing step. Our focus was to compare low-, medium-, and high-altitude polar orbits and quantify their visibility characteristics under the same terrain constraints.

1. Approach

We modeled how a lunar-orbiting augmentation satellite would appear from inside Shackleton by evaluating three circular polar orbits at different altitudes: a low polar orbit (LPO, 100 km), a medium polar orbit (MPO, 500 km), and a high polar orbit (HPO, 16,700 km). Although more advanced options such as the southern L2 NRHO are strong candidates for future navigation infrastructure, our goal here was to isolate how visibility inside Shackleton scales purely with orbital altitude. Circular polar orbits provide a clean and interpretable comparison without introducing additional dynamical effects.

Each orbit was propagated for one full period using poliastro's two-body Cowell formulation pol (2024b), and 200 uniformly spaced satellite states were sampled along each trajectory. These Moon-centered inertial position vectors were converted into Shackleton's local East–North–Up frame using standard ECEF-to-ENU geometry Soler and Eisemann (1994); **?**); European Space Agency (2012). Shackleton's coordinates were obtained from the USGS crater database Robbins (2019). Elevation and azimuth were computed from the resulting line-of-sight vectors Gongora-Torres et al. (2023).

To determine visibility, we used the DEM-derived horizon mask produced in the first stage of the project. This mask, based on LOLA topography Smith et al. (2016), provides the minimum elevation angle required for line-of-sight as a function of azimuth. For each satellite sample, the required elevation was interpolated from the mask, and visibility was assigned by comparing actual satellite elevation against the terrain-limited threshold. All visibility calculations were performed from the geometric center of Shackleton for consistency.

This framework in propagation, coordinate transformation, and horizon-mask filtering, was applied identically across all three orbits, providing a consistent basis for the comparative results that follow.

2. Results

To study how orbital altitude affects visibility inside Shackleton, we compared three circular polar orbits (LPO, MPO, HPO) and kept everything else fixed so the only variable changing was altitude. We aimed to get quantify how terrain geometry and orbit altitude interact. Table 2 summarizes the expected behavior across the orbits we selected.

Orbit type	Model	Altitude	Period	Polar / Shackleton visibility behavior
Low Circular Polar Orbit (LPO)	2-body	∼100 km	∼118 min	Very short visibility. Orbit is too close to the surface, so crater walls block most passes.
Medium Circular Polar Orbit (MPO)	2-body	∼500 km	∼158 min	Slightly longer visibility, but still heavily obstructed by crater topography.
High Circular Polar Orbit (HPO)	2-body	∼16,700 km	∼54 h	Long, continuous passes above the crater rim, with much less blockage from local terrain.
Near Rectilinear Halo Orbit (NRHO)	CR3BP	∼70,000 km	∼6–7 days	High-altitude south-pole passes give sustained Shackleton visibility. Selected for Nasa Gateway

Table 2: Comparison of candidate lunar orbits used in the trade study.

Although Equatorial Lunar Frozen Orbits (ELFOs) and other orbit families are usually proposed for lunar navigation constellations, we did not include them in this study because our goal was to isolate the relationship between altitude and terrain-induced visibility. ELFO dynamics are governed by different stability criteria and require CR3BP modeling, which would introduce additional effects unrelated to the altitude–visibility relationship we focused on.

Figure 8 shows the skyplots after projecting all satellite positions into Shackleton's local ENU frame and applying the DEM-derived horizon mask. Each orbit was sampled at 200 evenly spaced points. Since the orbits have very different periods, the angular spacing between these samples changes: the lower orbits move quickly in the sky and the samples spread out, while the high-altitude orbit moves slowly and appears almost continuous along a vertical line.

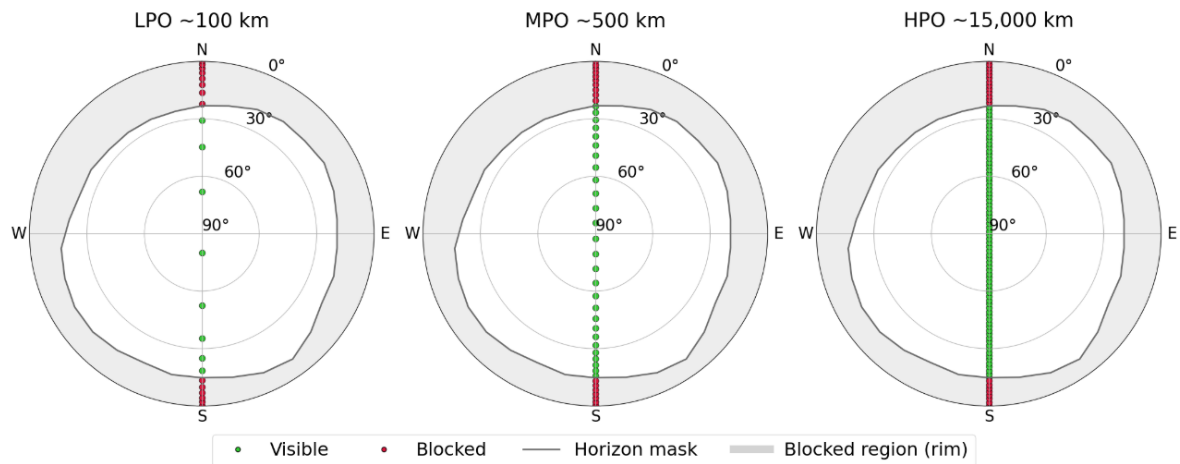


Figure 8: Shackleton skyplots for the low (LPO), medium (MPO), and high (HPO) circular polar orbits. Green points indicate line-of-sight visibility; red points are blocked by terrain.

To look at visibility over time, we plotted elevation versus sample index for each orbit (Figure 9). Since we fixed the sampling to 200 points per period, each orbit ends up with a different time spacing of about 0.6 minutes per sample for LPO, 0.8 minutes for MPO, and about 16 minutes for HPO. We interpolated the DEM horizon mask to get the required elevation at every azimuth, then compared it against the actual satellite elevation to determine when line-of-sight was available.

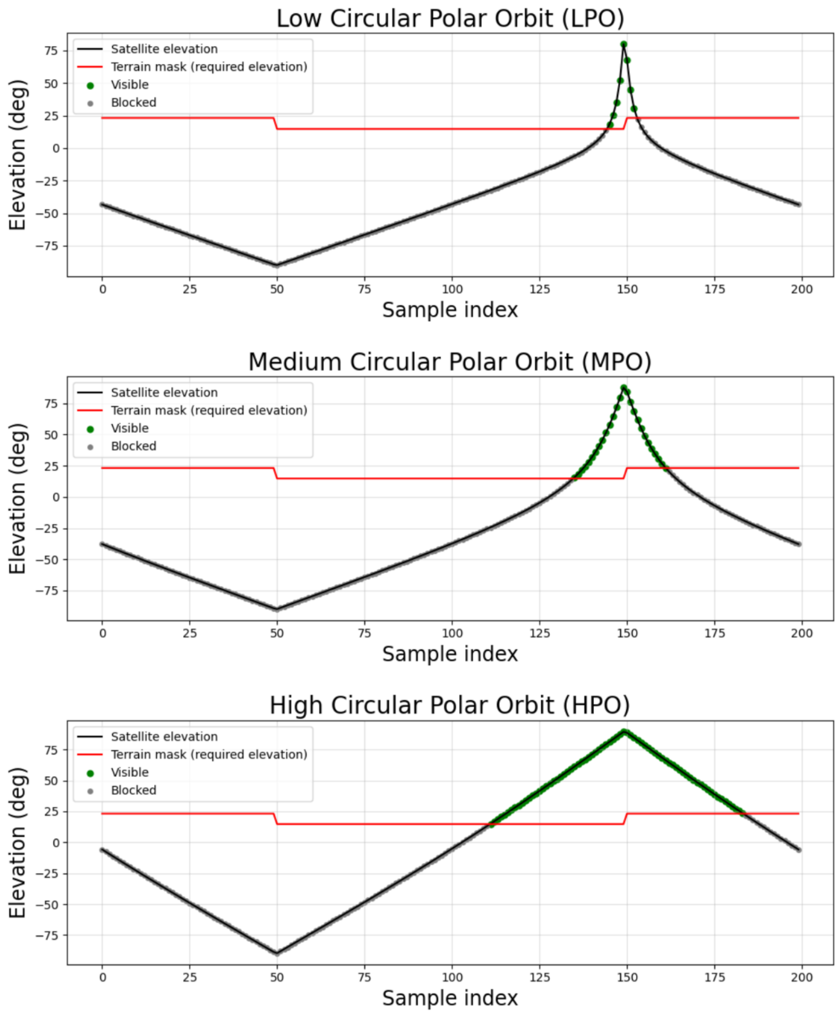


Figure 9: Elevation vs. sample index for LPO, MPO, and HPO. Visibility occurs when elevation exceeds the terrain-limited threshold.

Table 3 consolidates the visibility results. The trend is exactly what geometry predicts. The low orbit barely rises above the crater walls, so it only achieves 4.0% visibility with a short 4.74-minute window. The medium orbit does noticeably better at 13% and about 20 minutes of continuous access. The high orbit is in a different regime entirely—its altitude keeps it above the rim for long stretches of the orbit, giving 36.5% visibility and nearly 20 hours of uninterrupted coverage.

Orbit Type	**Altitude**	**Orbital Period**	**Time Between Samples (Δt)**	**Visibility**	**Max Continuous Visibility**
Low Circular Polar Orbit (LPO)	100 km	117.79 min	118 / 200 ≈ 0.6 min per sample	4.0%	4.74 min
Medium Circular Polar Orbit (MPO)	500 km	158.28 min	158 / 200 ≈ 0.8 min per sample	13.0%	20.68 min
High Circular Polar Orbit (HPO)	16,737 km	3238.46 min ∼54 hrs	3238 / 200 ≈ 16.2 min per sample	36.5%	1187.98 min ∼20 hrs

Table 3: Visibility metrics at Shackleton crater for three circular polar orbits.

Once the satellite reaches high enough altitude, local topography stops dominating the geometry. Visibility grows quickly with altitude, and the HPO case demonstrates how powerful this effect is. This same principle is part of the motivation behind south-pole NRHO architectures—high-altitude trajectories naturally maintain line-of-sight into deep polar craters.

VII. CONTRIBUTIONS OF TEAM MEMBERS

1. Henry's Work

During this project, Henry focused on the development and usage of the ray tracing tooling described in Section IV above. Library usage was minimal throughout the project, so the code dealing with geometric transformations, file input/output, graph representations, and more were all custom. Debugging and iterating on this code took the majority of the time and was necessary to provide the other team members with any relevant data product. Lastly, Henry contributed to the presentation and this write-up along with his teammates.

2. Pablo's Work

During the project, Pablo focused on the analysis of site error. For this, the analysis was initiated using the code and information from H2, Problem 3. This served to provide a way to calculate DOP and graph the H tables. After that, Pablo focused on graphing the elevation or horizon mask for each of the sites analyzed in the homework, exemplifying what an urban canyon behavior on the Moon would look like.

3. Sabrina's Work

Sabrina focused on the lunar satellite augmentation, where her goal was to quantify how a lunar orbiter would improve navigation by calculating PDOP values on the Shackleton crater. However, because there was no access to the minimum number of satellites needed to calculate PDOP inside the crater, she shifted her analysis into quantifying how the altitude of the lunar orbit impacts visibility inside craters to influence future constellation design. The main libraries used were Poliastro and PyMap3d for orbit propagation and 3D coordinate conversion respectively, and the rest of the code was original. Sabrina was also responsible for an extensive literature review, and she contributed to the presentation and report write-up along with her teammates.

VIII. CONCLUSION

In the end, this project was successful in developing tooling that could be used as a part of robotic mission design on the moon. Specifically, this tooling addresses the issue of localization, because while South Pole craters present several benefits such as water ice and rims of perpetual sunlight, localization is a challenge because of inconsistent or non-existent visibility of GNSS satellites. Ray tracing can be a valuable tool in assessing this satellite visibility by calculating the obstructions between a robotic explorer and the GNSS satellites around Earth. Furthermore, ray tracing can be used to calculate the horizon line for a given crater, which allows for the calculation of site-specific DOP values. As the final piece of analysis done in this project, the team looked at the possibility of adding a lunar satellite for augmented localization, comparing different potential orbits based on the calculated horizon line. Thus, this project successfully demonstrates the usage of a ray-tracing-based framework for the assessment of lunar GNSS-based localization techniques.

IX. FUTURE DIRECTION

As seen throughout this project, there are several areas where the analysis and simulations could be improved. Our initial topography-based visibility analysis of GNSS signals on the lunar surface was limited by resolution and compute constraints, so the next step is to use higher-fidelity DEMs to generate more accurate horizon masks and satellite visibility maps.

For the site error analysis, we relied on CSV snapshots of Earth-orbiting satellites at a single epoch and evaluated visibility only at the crater centers. While this was sufficient to illustrate how elevation masks affect DOP, a proper study would track satellite positions over time relative to the rotating and orbiting Moon and evaluate visibility at multiple points within each crater. Viewpoint changes, such as standing closer to crater walls, would also alter the effective elevation mask and block additional satellites.

Finally, in the lunar-orbiting augmentation study, we restricted ourselves to two-body circular polar orbits to stay within the project scope. A more complete investigation would include the full catalogue of feasible lunar orbits, especially ELFOs, NRHOs and other orbit families, and would compare them over much longer time to determine how many satellites are needed to meaningfully improve DOP. A transient, time-varying analysis would be required to evaluate both Earth-GNSS geometry and the new lunar satellites simultaneously. Additionally, as visibility inside lunar craters near the rim gives wider visibility, we will focus our future work on how access degrades from rim to floor.

X. CODE

All code used in this research can be found on `https://github.com/sabrinanicacio/GPS-AA272`.

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